# Sharp pointwise bounds for Lebesgue functions

Qiyuan Gu
University of Chicago
phoenix1203@uchicago.edu

**Abstract**

For each $n$, let $\lambda_n$ be the Lebesgue function for polynomial interpolation at an arbitrary set of $n$ distinct nodes in $[-1,1]$. We prove that the points $x \in (-1,1)$ for which $\lambda_n(x) > (2/\pi)\log n - C(x)$ infinitely often, with some finite $C(x)$, form a dense set. We also prove $\limsup_{n\to\infty} \lambda_n(x)/\log n \geq 2/\pi$ for almost every $x \in (-1,1)$. The two proofs use derivative-jump energies and positive Cauchy transforms, respectively. Both are formalized in Lean 4.

## 1 Introduction

For each $n \geq 1$, let

$$-1 \leq x_{1,n} < \cdots < x_{n,n} \leq 1.$$

Define the node polynomial, the cardinal polynomials, and the Lebesgue function by

$$P_n(x) = \prod_{k=1}^{n} (x - x_{k,n}),$$

$$\ell_{k,n}(x) = \frac{P_n(x)}{(x - x_{k,n})P_n'(x_{k,n})}, \qquad \lambda_n(x) = \sum_{k=1}^{n} |\ell_{k,n}(x)|.$$

The rows of the array need not be nested.

**Theorem 1.** *For every triangular array of distinct nodes as above, the following statements hold.*

*(i) The set of points $x \in (-1,1)$ for which there is a finite constant $C(x)$ such that*

$$\lambda_n(x) > \frac{2}{\pi}\log n - C(x) \tag{1.1}$$

*for infinitely many $n$ is dense in $(-1,1)$.*

*(ii) For Lebesgue almost every $x \in (-1,1)$,*

$$\limsup_{n\to\infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}. \tag{1.2}$$

Theorem 1 answers the two questions in Erdős Problem 1132 [2], with a point-dependent constant in part (i). An absolute constant cannot replace $C(x)$: for every $M > 0$, there is an array such that, at every $x \in (-1,1)$,

$$\lambda_n(x) \leq \frac{2}{\pi}\log n - M \qquad (n \geq n_0(x)).$$

This construction is given in [10, Theorem 1].

## 1.1 Background

Faber [9] proved the order $\log n$ for the global Lebesgue constant, Bernstein [1] obtained the asymptotically sharp leading constant $2/\pi$, Erdős and Turán [6] improved the error to $O(\log\log n)$, Erdős [4] to $O(1)$, and Vértesi [12, Corollary 1] refined the asymptotics through the optimal constant term to an error of $O((\log\log n/\log n)^2)$. See [11, Section 1.3] and [3] for further historical discussion.

For triangular arrays, Erdős [5, p. 68, equation (7)] stated that the set of points $x$ satisfying

$$\limsup_{n\to\infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}$$

is everywhere dense, deducing this from an interval estimate whose proof he described as unpublished. He asked whether the normalized bound holds almost everywhere, and separately whether the logarithmic bound holds infinitely often at a point with a bounded additive loss. Erdős and Vértesi [7] proved almost-everywhere divergence of interpolants of a suitable continuous function for every array. Their quantitative estimate [8, Theorem 2.1] gives logarithmic growth outside a set of arbitrarily small measure, with a positive coefficient depending on that measure. In particular,

$$\limsup_{n\to\infty} \frac{\lambda_n(x)}{\log n} > 0 \quad \text{for almost every } x.$$

Theorem 1(ii) gives the sharp coefficient $2/\pi$.

Tao [11, Theorem 1.10(i)] proved that every fixed nondegenerate interval $I \subset [-1,1]$ satisfies

$$\sup_{x\in I} \lambda_n(x) \geq \frac{2}{\pi}\log n - O_I(1).$$

His Corollary 1.11 and Remark 1.12 give, for every prescribed $\omega(n) \to \infty$, a dense, indeed comeager, set of points $x$ for which

$$\lambda_n(x) \geq \frac{2}{\pi}\log n - \omega(n)$$

infinitely often. Theorem 1(i) replaces $\omega(n)$ by a finite constant depending on the point, on a dense set.

## 1.2 Proof outline

The proofs of Theorem 1(i) and (ii) are independent. We begin with the self-contained proof of part (ii) in Sections 2–3: cancellation in positive Cauchy transforms and a harmonic-measure identity rule out a common low set of positive measure. For part (i), Sections 4–6 use Tao's local estimates [11, Theorem 4.1(ii)], together with Riesz's differentiation formula, to turn large derivative jumps into nearby points with a bounded additive loss. A Riesz energy bound and a recurrence argument give a positive-measure set of such points, and Baire's theorem gives density. Section 7 gives a positive-measure consequence.

Throughout the paper, all logarithms are natural and $|E|$ denotes Lebesgue measure. We write $c_0 = 2/\pi$, and suppress the row index when discussing a single row. Constants in local estimates may depend on the fixed intervals and test functions, but not on $n$ or on interpolation data bounded by one. The notation $J \Subset U$ means that $J$ is a compact subset of the open set $U$.

## 2 Positive Cauchy transforms

Suppress the row index and put

$$S = \sum_k \frac{1}{|P'(x_k)|}, \qquad \nu_k = \frac{1}{S|P'(x_k)|}, \qquad \sigma_k = \operatorname{sgn} P'(x_k).$$

Define

$$m(z) = \sum_k \frac{\nu_k}{z - x_k}, \qquad g(z) = \sum_k \frac{\sigma_k \nu_k}{z - x_k} = \frac{1}{SP(z)}. \tag{2.1}$$

The last identity is the partial fraction formula for $1/P$, whose expansion at infinity gives

$$\sum_k \sigma_k \nu_k x_k^j = 0 \qquad (0 \le j \le n-2). \tag{2.2}$$

For $n \ge 2$, both $m + g$ and $m - g$ are Cauchy transforms of positive probability measures: their masses at $x_k$ are $(1 \pm \sigma_k)\nu_k$.

For real $x$ away from the nodes, write

$$H(x) = \sum_k \frac{\nu_k}{|x - x_k|}, \qquad \gamma_s(x) = -\Im m(x + is), \qquad p_s(t) = \frac{s}{\pi(t^2 + s^2)}.$$

Then

$$\lambda_n(x) = \frac{H(x)}{|g(x)|}, \qquad H(x) = \frac{2}{\pi} \int_0^\infty \gamma_s(x)\, \frac{ds}{s}, \tag{2.3}$$

and the Poisson semigroup gives

$$\gamma_s = p_{s-h} * \gamma_h \qquad (s > h). \tag{2.4}$$

**Lemma 2.** *For every row with $n \ge 2$, every $0 < h \le 1$, and every $x \in [-1, 1]$,*

$$\frac{|g(x + ih)|}{\gamma_h(x)} \le \frac{4 + h^2}{h^2 \sinh((n-1)h/2)}. \tag{2.5}$$

*Consequently, if $0 < a < 1$ and $h = n^{-a}$, then*

$$\sup_{x \in [-1,1]} \frac{|g(x + ih)|}{\gamma_h(x)} \longrightarrow 0. \tag{2.6}$$

*Proof.* Set $z = x + ih$ and $p = T_{n-1}$, the Chebyshev polynomial of the first kind. The polynomial $(p(t) - p(z))/(t - z)$ has degree at most $n - 2$, so (2.2) gives

$$p(z)g(z) = \sum_k \frac{\sigma_k \nu_k p(x_k)}{z - x_k}, \qquad |p(z)g(z)| \le \frac{1}{h},$$

using $|p(x_k)| \le 1$ and $\sum_k \nu_k = 1$. Write $z = \cos(u + iv)$, with $u, v \in \mathbb{R}$. Then $h \le \sinh |v|$, so $|v| \ge \operatorname{arsinh} h \ge h/2$. The identity $T_{n-1}(\cos w) = \cos((n-1)w)$ yields

$$|p(z)|^2 = \cos^2((n-1)u) + \sinh^2((n-1)v) \ge \sinh^2((n-1)h/2).$$

Finally, since the nodes and $x$ lie in $[-1, 1]$,

$$\frac{h}{4 + h^2} \le \gamma_h(x) = \sum_k \frac{\nu_k h}{(x - x_k)^2 + h^2}.$$

Combining these bounds proves (2.5). For $h = n^{-a}$, its right-hand side tends to zero: the hyperbolic sine grows exponentially in $n^{1-a}$, while $h^{-2}$ grows polynomially. This proves (2.6). □

**Lemma 3.** *Let $F$ be the Cauchy transform of a positive probability measure with finite support. For every bounded real interval $S_0$, every $x \in \mathbb{R}$, and every $h > 0$,*

$$\int_{\mathbb{R}} p_h(x-y)\, \mathbf{1}_{\{F(y)\in S_0\}}\, dy = \frac{1}{\pi}\int_{S_0} \frac{-\Im F(x+ih)}{(t-\Re F(x+ih))^2 + (\Im F(x+ih))^2}\, dt. \tag{2.7}$$

*Proof.* For $\Im w < 0$, define

$$B(w) = \frac{1}{\pi}\int_{S_0} \frac{-\Im w}{(t-\Re w)^2 + (\Im w)^2}\, dt.$$

The function $B$ is harmonic and lies between zero and one. Since $F$ maps the upper half-plane into the lower half-plane, $B \circ F$ is bounded and harmonic. Away from the poles of $F$ and the preimages of the endpoints of $S_0$, its boundary value is $\mathbf{1}_{\{F(y)\in S_0\}}$. These exceptions form a finite set: $F$ is a nonconstant rational function. The Poisson representation of the bounded harmonic function $B \circ F$ now gives (2.7). □

# 3 The almost-everywhere bound

*Proof of Theorem 1(ii).* The failure set is the union, over rational $0 < c < c_0$ and integers $N \geq 2$, of the measurable sets

$$\{y \in (-1,1) : \lambda_n(y) \leq c \log n \text{ for every } n \geq N\}.$$

If the assertion fails, one of these sets $E$ has positive measure. For its fixed $c, N$, we therefore have

$$\lambda_n(y) \leq c\log n \qquad (y \in E,\ n \geq N). \tag{3.1}$$

Choose constants

$$K > 1, \qquad 0 < b < a < 1, \qquad \frac{2b}{\pi c K} > 1,$$

which is possible because $c < 2/\pi$. Set

$$h = n^{-a}, \qquad \eta = n^{-b}, \qquad H^\eta(x) = \frac{2}{\pi}\int_\eta^1 \gamma_s(x)\, \frac{ds}{s}.$$

By (2.3), $H^\eta \leq H$. For $s > 0$ and real $x, y, t$, the triangle inequality gives

$$|y + is - t| \leq |x + is - t| + |x - y| \leq \left(1 + \frac{|x-y|}{s}\right)|x + is - t|.$$

Taking reciprocal squares and summing against the positive masses yields $\gamma_s(x) \leq (1 + |x - y|/s)^2\gamma_s(y)$. Integrating over $\eta \leq s \leq 1$, we obtain

$$H^\eta(x) \leq \left(1 + \frac{|x-y|}{\eta}\right)^2 H^\eta(y). \tag{3.2}$$

Fix $x \in E$, and write

$$u = \Re m(x+ih), \qquad \gamma = \gamma_h(x), \qquad S_x = [u - K\gamma, u + K\gamma].$$

By (2.6) and (2.7), each of the two events

$$m(y) + g(y) \in S_x, \qquad m(y) - g(y) \in S_x$$

has probability

$$p_K + o(1), \qquad p_K = \frac{2}{\pi}\arctan K > \frac{1}{2},$$

under the Poisson probability $p_h(x-y)\,dy$. The error is uniform in $x \in E$: for $F = m \pm g$, (2.6) gives $\Re F(x+ih) = u + o(\gamma)$ and $-\Im F(x+ih) = \gamma(1+o(1))$. Substituting $t = u + \gamma s$ in (2.7) therefore gives $\pi^{-1}\int_{-K}^{K}(1+s^2)^{-1}\,ds + o(1) = p_K + o(1)$. The two events have intersection probability at least

$$q_K + o(1), \qquad q_K = 2p_K - 1 > 0. \tag{3.3}$$

Choose $R$ so that the Poisson mass of $|y-x| > Rh$ is less than $q_K/8$, and let

$$E_n = \{x \in E : (p_h * \mathbf{1}_{\mathbb{R}\setminus E})(x) < q_K/8\}.$$

The approximate identity theorem gives $|E \setminus E_n| \to 0$. For all sufficiently large $n$, (3.3) is greater than $q_K/2$. For $x \in E_n$, the two excluded sets $\mathbb{R} \setminus E$ and $\{|y-x| > Rh\}$ have total Poisson mass less than $q_K/4$. Hence there exists $y \in E$ with $|y - x| \le Rh$ and both events true. We may take $y$ away from the nodes. Since $m(y) \pm g(y)$ both lie in an interval of length $2K\gamma_h(x)$,

$$|g(y)| \le K\gamma_h(x).$$

Equations (2.3) and (3.1) give

$$H^\eta(y) \le H(y) = \lambda_n(y)|g(y)| \le cK(\log n)\gamma_h(x).$$

Together with (3.2), this yields

$$H^\eta(x) \le (1 + Rh/\eta)^2 H^\eta(y) \le cK(1+Rh/\eta)^2(\log n)\gamma_h(x) \qquad (x \in E_n). \tag{3.4}$$

Let $\mathcal{K}_n$ be convolution with the symmetric probability kernel

$$k_n(t) = \frac{1}{b\log n}\int_\eta^1 p_{s-h}(t)\,\frac{ds}{s}.$$

Its integral is one, since $\int_\eta^1 ds/s = b\log n$. Equation (2.4) then shows that

$$H^\eta = \frac{2b}{\pi}(\log n)\mathcal{K}_n\gamma_h.$$

Thus (3.4) becomes

$$\frac{\mathcal{K}_n\gamma_h(x)}{\gamma_h(x)} \le \frac{1}{A_n} \quad (x \in E_n), \qquad A_n = \frac{2b}{\pi c K}(1 + Rh/\eta)^{-2} \longrightarrow A > 1. \tag{3.5}$$

Integrate (3.5) over $E_n$, discard the contribution from $y \notin E_n$, and symmetrize. Positivity of $\gamma_h$, symmetry of $k_n$, and $(r + r^{-1})/2 \ge 1$ give

$$\frac{|E_n|}{A_n} \ge \iint_{E_n\times E_n} k_n(x-y)\frac{\gamma_h(y)}{\gamma_h(x)}\,dy\,dx \ge \iint_{E_n\times E_n} k_n(x-y)\,dy\,dx. \tag{3.6}$$

The kernels $k_n$ form an approximate identity. In fact, for every fixed $\delta > 0$,

$$\int_{|t|>\delta} k_n(t)\,dt \le \frac{C}{b\log n}\int_\eta^1 \min(1, s/\delta)\,\frac{ds}{s} \longrightarrow 0.$$

It follows that

$$\iint_{E\times E} k_n(x-y)\,dy\,dx \longrightarrow |E|.$$

Replacing $E$ by $E_n$ changes this integral by at most $2|E \setminus E_n|$. Taking limits in (3.6) therefore yields

$$\frac{|E|}{A} \ge |E|,$$

contrary to $|E| > 0$ and $A > 1$. This proves the assertion. $\square$

## 4 A local differentiation formula

We use the following local form of Riesz's differentiation formula and its near-equality consequence.

**Lemma 4.** *Suppose that $T \geq 2\pi$ and $F$ is holomorphic in a neighborhood of the disk $\{|z| \leq T\}$, is real on its real diameter, and satisfies, throughout this disk,*

$$|F(u+iv)| \leq e^{|v|}.$$

*Then*

$$|F'(0)| \leq 1 + O(T^{-1}). \tag{4.1}$$

*If $F'(0) \geq 1-\delta$, where $\delta \geq 0$, then*

$$F(\pi/2) \geq 1 - \frac{\pi^2}{4}\delta - O(T^{-1}). \tag{4.2}$$

*The constants are absolute.*

*Proof.* Choose $R = m\pi$ with $T/2 \leq R < T$, and integrate

$$\frac{F(z)}{z^2 \cos z}$$

around $|z| = R$. On this circle, $e^{|\Im z|} \leq 8|\cos z|$. Indeed, for $|v| \geq 1$, use $|\cos(u+iv)| \geq \sinh|v| \geq e^{|v|}/4$. For $|v| \leq 1$, put $d = R - |u| = v^2/(R+|u|) \leq 1/2$; then $|\cos u| = |\cos d| \geq 1/2$, while $e^{|v|} \leq e < 3$. The contour integral, divided by $2\pi i$, has absolute value at most $8/R$. Taking residues at zero and at $t_j = (j+\frac{1}{2})\pi$ gives

$$F'(0) = \sum_{|t_j|<R} \frac{(-1)^j F(t_j)}{t_j^2} + O(T^{-1}). \tag{4.3}$$

Applying the same formula to $F(z) = \sin z$ gives

$$\sum_{|t_j|<R} \frac{1}{t_j^2} = 1 + O(T^{-1}).$$

Together with $|F(t_j)| \leq 1$, equation (4.1) follows. Isolating the term $t_0 = \pi/2$ in (4.3) gives

$$1 - \delta \leq \frac{4}{\pi^2}F(\pi/2) + 1 - \frac{4}{\pi^2} + O(T^{-1}),$$

which proves (4.2). □

We apply this formula to interpolants with bounded nodal data. We first state the local estimates in the form used here, with the normalization:

$$S_n = \sum_k \frac{1}{|P_n'(x_{k,n})|}, \qquad \alpha_n = \frac{1}{n}\log S_n, \qquad U_n(z) = \frac{1}{n}\log\frac{1}{|P_n(z)|}.$$

The following proposition is Tao's local estimate [11, Theorem 4.1(ii), equations (4.6)–(4.7)]; the normalization and density are those of his equations (4.11) and (4.15). We specialize his polynomial bound $\sup_I \lambda_n \ll n^{O(1)}$ to $\sup_I \lambda_n \leq n$.

**Proposition 5** (Local estimates)**.** *Let $I \Subset (-1,1)$ be a fixed nondegenerate closed interval, and suppose that $\sup_I \lambda_n \le n$ for all sufficiently large $n$. Define the approximate local node density by*

$$\rho_n(x) = -\frac{1}{\pi^2}\int_{\mathbb{R}\setminus I} \frac{U_n(y)-\alpha_n}{(y-x)^2}\,dy.$$

*For each fixed compact interval $I' \Subset \operatorname{int} I$, the following hold uniformly for all sufficiently large $n$:*

*(i) The densities are bounded above and below by positive constants and have uniformly bounded Lipschitz norms:*

$$\rho_n(x) \asymp_{I'} 1, \qquad |\rho_n(x)-\rho_n(x_0)| \ll_{I'} |x-x_0| \qquad (x, x_0 \in I').$$

*(ii) For $x \in I'$ and $\eta \ge \log n/n$,*

$$U_n(x+i\eta) = \alpha_n - \pi\eta\rho_n(x) + O_{I'}\left(\frac{\log n}{n} + \eta^3\right). \tag{4.4}$$

**Lemma 6** (Local quadrature)**.** *Under the hypotheses of Proposition 5, fix a compact interval $I' \Subset \operatorname{int} I$. For globally Lipschitz functions $\varphi_n$ supported in $I'$, with uniformly bounded supremum norms and Lipschitz constants,*

$$\frac{1}{n}\sum_k \varphi_n(x_{k,n}) = \int \varphi_n(x)\rho_n(x)\,dx + O(n^{-1/4}). \tag{4.5}$$

*Proof.* Let $\mu_n = n^{-1}\sum_k \delta_{x_{k,n}}$ and $Q_h = h^{-1}\int_h^{2h} p_t\,dt$, where $p_t(y) = t/(\pi(y^2+t^2))$. This is a probability kernel with $\int_{|y|>\sqrt{h}} Q_h(y)\,dy = O(\sqrt{h})$, so $\|Q_h * \varphi_n - \varphi_n\|_\infty = O(\sqrt{h})$. The Poisson representation gives

$$(Q_h * \mu_n)(x) = \frac{U_n(x+ih) - U_n(x+2ih)}{\pi h} = \rho_n(x) + O\left(\frac{\log n}{nh} + h^2\right) \qquad (x \in I').$$

Integrating against $\varphi_n$ and using Fubini therefore gives an error $O(\sqrt{h} + \log n/(nh) + h^2)$ in (4.5). Taking $h = n^{-1/2}$ proves the claim. ☐

**Lemma 7.** *Let $I \Subset (-1,1)$ be a fixed nondegenerate closed interval. Suppose that, for all sufficiently large $n$,*

$$\lambda_n(x) \le \Lambda_n := c_0 \log n + C \qquad (x \in I), \tag{4.6}$$

*where $C$ is fixed. This bound gives the hypothesis of Proposition 5; let $\rho_n$ be its densities. For every fixed interval $J \Subset \operatorname{int} I$ and every real interpolant*

$$q(x) = \sum_{k=1}^n a_k \ell_{k,n}(x), \qquad |a_k| \le 1,$$

*uniformly for $x \in J$,*

$$|q'(x)| \le \pi n \rho_n(x)(\Lambda_n + o(1)), \qquad |q''(x)| \le C_J n^2 \log n. \tag{4.7}$$

*Moreover, for every fixed $D$, if*

$$\frac{|q'(x)|}{\pi n \rho_n(x)} \ge c_0 \log n - D, \tag{4.8}$$

*then there is $y = x + O_J(n^{-1})$ such that*

$$\lambda_n(y) \ge c_0 \log n - D', \tag{4.9}$$

*where $D'$ depends on $I, J, C, D$, but not on $n, x, q$.*

*Proof.* Fix $I' \Subset \operatorname{int} I$ with $J \Subset \operatorname{int} I'$. For $\Re z \in I'$ and $\Im z = \eta \geq \log n/n$, the interpolation formula and (4.4) give

$$|q(z)| \leq \frac{S_n|P_n(z)|}{\eta} \leq \frac{1}{\eta}\exp\Big(\pi n\eta\rho_n(\Re z) + O(\log n + n\eta^3)\Big). \tag{4.10}$$

Fix $x \in J$, and put

$$L = n^{-1/5}, \qquad H = n^{-2/5}, \qquad \omega = \pi n\rho_n(x)(1 + n^{-1/10}).$$

On the upper edge of $\{|\Re z - x| \leq L,\ 0 \leq \Im z \leq H\}$, we have $|q(z)| \leq \Lambda_n e^{\omega H}$: $\omega H$ exceeds $\pi n H\rho_n(x)$ by $\pi\rho_n(x)n^{1/2}$, which dominates the $O(n^{2/5})$ from the Lipschitz variation of $\rho_n$ and the remaining $O(\log n)$ terms in (4.10). On the lower edge $|q| \leq \Lambda_n$; the Bernstein–Walsh bound gives $|q| \leq \Lambda_n \exp(O_I(n))$ on the vertical sides. For $f(z) = \Lambda_n^{-1}q(z)e^{i\omega(z-x)}$, the horizontal edges have $|f| \leq 1$, and the vertical edges have $|f| \leq e^B$ with $B = O_I(n)$. Set

$$\Psi(z) = 2B\left[e^{(z-x-L)/H} + e^{(-z+x-L)/H}\right].$$

Its real part is nonnegative on the horizontal edges and at least $B$ on the vertical edges, since $\cos(v/H) \geq 1/2$ for $0 \leq v \leq H$. The maximum modulus principle gives $|f(z)e^{-\Psi(z)}| \leq 1$. At distance at least $L/2$ from the vertical sides, the resulting error factor is $\exp(O(ne^{-L/(2H)})) = 1 + O(n^{-10})$. Thus, also reflecting across the real axis,

$$|q(x+u+iv)| \leq \Lambda_n(1+n^{-10})e^{\omega|v|} \qquad (|u| \leq L/2,\ |v| \leq H). \tag{4.11}$$

The rescaled function

$$F(t) = \frac{q(x+t/\omega)}{\Lambda_n(1+n^{-10})}$$

satisfies Lemma 4 on a disk of radius $T \asymp_J n^{3/5}$. By (4.1),

$$|q'(x)| = \omega\Lambda_n(1+n^{-10})|F'(0)| \leq \pi n\rho_n(x)\Lambda_n(1+n^{-1/10})(1+O(n^{-3/5})).$$

Since $\Lambda_n n^{-1/10} = o(1)$, this gives the first bound in (4.7); Cauchy's estimate on a disk of radius comparable to $n^{-1}$ gives the second. Under (4.8), change the sign of $q$ if necessary. For sufficiently large $n$,

$$F'(0) \geq \frac{c_0\log n - D}{\Lambda_n(1+n^{-1/10})(1+n^{-10})} \geq 1 - \frac{C_1}{\Lambda_n}, \qquad C_1 = 1 + \max\{C+D, 0\}.$$

Equation (4.2) then gives $|q(x+\pi/(2\omega))| \geq \Lambda_n - C_*$. Since $|q(y)| \leq \lambda_n(y)$ on the real axis and $\omega \asymp_J n$, this proves (4.9) with the asserted displacement. $\square$

## 5 Derivative jumps and a Riesz energy

At each internal interpolation node, write

$$\Delta_k = \lambda_n'(x_{k,n}+) - \lambda_n'(x_{k,n}-).$$

At the first and last nodes, use the exterior polynomial pieces. Set

$$\nu_k = \frac{1}{S_n|P_n'(x_{k,n})|}.$$

The absolute values of all cardinal polynomials other than $\ell_{k,n}$ have a cusp at $x_{k,n}$, whereas $\ell_{k,n}$ is positive near that node. Using $\ell_{j,n}(x) = P_n(x)/((x - x_{j,n})P_n'(x_{j,n}))$ and $P_n(x_{k,n}) = 0$ to differentiate at $x_{k,n}$, we obtain

$$\Delta_k = 2\sum_{j\neq k} |\ell_{j,n}'(x_{k,n})| = 2\sum_{j\neq k} \frac{\nu_j}{\nu_k |x_{k,n} - x_{j,n}|}. \tag{5.1}$$

**Lemma 8.** *Under the hypotheses of Lemma 7, choose a nonzero nonnegative smooth function $f$ compactly supported in* $\operatorname{int} I$*. Then*

$$\sum_k \frac{f(x_{k,n})^2}{n\rho_n(x_{k,n})} \frac{\Delta_k}{2\pi n\rho_n(x_{k,n})} \geq c_0 \log n \int f(x)^2\, dx - O(1). \tag{5.2}$$

*Terms with $f(x_{k,n}) = 0$ are taken to be zero; $\rho_n$ is only needed on a fixed interior neighborhood of the support of $f$.*

*Proof.* Put $a_k = f(x_{k,n})/\rho_n(x_{k,n})$, with $a_k = 0$ outside the support of $f$, and define

$$\mu_n = \frac{1}{n}\sum_k a_k \delta_{x_{k,n}}, \qquad K_\varepsilon(t) = \frac{1}{\sqrt{t^2+\varepsilon^2}}.$$

The kernel $K_\varepsilon$ is positive definite, since

$$K_\varepsilon(t) = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-s\varepsilon^2} e^{-st^2}\, ds$$

is a positive mixture of Gaussian kernels. Write

$$\mathcal{E}_\varepsilon(\sigma,\tau) = \iint K_\varepsilon(x-y)\, d\sigma(x)\, d\tau(y)$$

for its bilinear energy. Positivity applied to $\mu_n - f(x)\,dx$ gives

$$\mathcal{E}_\varepsilon(\mu_n,\mu_n) \geq 2\mathcal{E}_\varepsilon(\mu_n, f\,dx) - \mathcal{E}_\varepsilon(f\,dx, f\,dx). \tag{5.3}$$

Take $\varepsilon = 1/n$. Subtracting $f(x)$ in the integral over $|t| \leq 1$, and using the Lipschitz bound for $f$, gives the uniform estimate

$$(K_{1/n} * f)(x) = 2\log n\, f(x) + O_f(1). \tag{5.4}$$

The mass of $\mu_n$ is bounded. Also, by (4.5),

$$\int f\, d\mu_n = \frac{1}{n}\sum_k \frac{f(x_{k,n})^2}{\rho_n(x_{k,n})} = \int f^2 + O(n^{-1/4}). \tag{5.5}$$

Again by (5.4),

$$\mathcal{E}_{1/n}(f\,dx, f\,dx) = 2\log n \int f^2 + O(1).$$

Together with (5.3)–(5.5), this yields

$$\mathcal{E}_{1/n}(\mu_n,\mu_n) \geq 2\log n \int f^2 - O(1). \tag{5.6}$$

The diagonal part of this energy is $n^{-1}\sum_k a_k^2 = O(1)$. Since $K_{1/n}(t) \le 1/|t|$ for $t \ne 0$,

$$\frac{1}{n^2}\sum_{k\ne j}\frac{a_k a_j}{|x_{k,n}-x_{j,n}|} \ge 2\log n \int f^2 - O(1). \tag{5.7}$$

Finally,

$$a_k^2\frac{\nu_j}{\nu_k} + a_j^2\frac{\nu_k}{\nu_j} \ge 2a_k a_j.$$

By (5.1), the left side of (5.2) is

$$\frac{1}{\pi n^2}\sum_{k\ne j}\frac{a_k^2\nu_j}{\nu_k|x_{k,n}-x_{j,n}|}.$$

Pairing the terms indexed by $(k,j)$ and $(j,k)$, and then applying (5.7), bounds this expression below by

$$\frac{1}{\pi n^2}\sum_{k\ne j}\frac{a_k a_j}{|x_{k,n}-x_{j,n}|} \ge \frac{2}{\pi}\log n \int f^2 - O(1).$$

This proves (5.2). □

# 6 A dense set of points with bounded additive loss

**Proposition 9.** *Assume the local upper bound* (4.6) *on a fixed nondegenerate interval* $I \Subset (-1,1)$. *There are a constant* $D$ *and a measurable set* $G \subset \operatorname{int} I$, *with* $|G| > 0$, *such that every* $x \in G$ *satisfies*

$$\lambda_n(x) > c_0 \log n - D$$

*for infinitely many* $n$.

*Proof.* Choose compact intervals $J_0 \Subset J_1 \Subset \operatorname{int} I$, and choose $f$ as in Lemma 8 with $\operatorname{supp} f \subset J_0$. All constants below are independent of $n$: after $I, C, J_0, J_1, f$ are fixed, we choose $D$, then $\kappa, D_1, d, \kappa_1, d_1$, and finally $b, D_2$. Let

$$w_k = \frac{f(x_{k,n})^2}{n\rho_n(x_{k,n})}, \qquad \widehat{\Delta}_k = \frac{\Delta_k}{2\pi n\rho_n(x_{k,n})}, \qquad W = \int f^2 > 0.$$

By (4.5),

$$\sum_k w_k = W + O(n^{-1/4}), \qquad 0 \le w_k \le C_f/n. \tag{6.1}$$

On each interval between successive nodes, $\lambda_n$ equals an interpolant with nodal data in $\{-1,1\}$. Both one-sided derivatives at a node therefore satisfy (4.7), so

$$0 \le \widehat{\Delta}_k \le \Lambda_n + o(1) \tag{6.2}$$

uniformly on the support of $f$. Lemma 8, (6.1), and (6.2) show that

$$\sum_k w_k(\Lambda_n + 1 - \widehat{\Delta}_k) = O(1), \tag{6.3}$$

with nonnegative summands for all sufficiently large $n$. On a node with $\widehat{\Delta}_k < c_0 \log n - D$, the factor in (6.3) is greater than $C + 1 + D$. Hence, by choosing $D$ sufficiently large, such nodes carry at most $W/2$ of the weights. Since (6.1) has total weight $W + o(1)$, for all sufficiently large $n$ the nodes in the support of $f$ satisfying

$$\widehat{\Delta}_k \geq c_0 \log n - D \tag{6.4}$$

carry at least $W/4$ of the weights. In particular, there are at least $\kappa n$ such nodes, with $\kappa > 0$ independent of $n$. Discarding the first and last nodes does not change this conclusion after reducing $\kappa$.

At a node satisfying (6.4), at least one adjacent polynomial piece $q$ has

$$|q'(x_{k,n})| \geq \pi n \rho_n(x_{k,n})(c_0 \log n - D).$$

Lemma 7 gives a point

$$y_k = x_{k,n} + O(n^{-1}), \qquad \lambda_n(y_k) \geq c_0 \log n - D_1. \tag{6.5}$$

The same polynomial piece $q$ has value 1 at both endpoints of the corresponding gap. Rolle's theorem and (4.7) imply that this gap has length at least $d/n$ for some fixed $d > 0$ when the gap is contained in $J_1$. If it is not contained in $J_1$, its length is bounded below by $\operatorname{dist}(J_0, \mathbb{R} \setminus J_1) > 0$.

An interval of length less than $d/n$ contains at most two such nodes, so every interval of length $O(n^{-1})$ contains only a bounded number of the nodes in (6.4). By (6.5), the same holds, with multiplicities, for the points $y_k$. Hence one can select at least $\kappa_1 n$ of them, separated by $d_1/n$, for fixed $\kappa_1, d_1 > 0$.

Equation (4.7), applied to every polynomial piece, also shows that $\lambda_n$ is $C_I n \log n$-Lipschitz on a fixed compact interior interval containing these points. Choose a fixed sufficiently small $b > 0$ and $D_2 > D_1 + C_I b$, and let $A_n$ be the union of the intervals of radius

$$r_n = \frac{b}{n \log n}$$

centered at the selected points. For all sufficiently large $n$, these intervals are disjoint, lie in a fixed compact interior interval, and satisfy

$$\lambda_n(x) > c_0 \log n - D_2 \quad (x \in A_n), \qquad \frac{c}{\log n} \leq |A_n| \leq \frac{C_2}{\log n}. \tag{6.6}$$

The separation of the centers in the $m$-th row implies, for every interval $B$,

$$|A_m \cap B| \leq C_3 \left( \frac{|B|}{\log m} + \frac{1}{m \log m} \right).$$

Summing this inequality over the at most $n$ components of $A_n$ gives, for $m > n$,

$$|A_n \cap A_m| \leq C_3 \left( \frac{|A_n|}{\log m} + \frac{n}{m \log m} \right). \tag{6.7}$$

Put $B_j = A_{2^j}$. Equations (6.6) and (6.7) give

$$|B_j| \asymp j^{-1}, \qquad |B_j \cap B_k| \leq C_4 \left( \frac{1}{jk} + \frac{2^{j-k}}{k} \right) \quad (j < k). \tag{6.8}$$

For every fixed $j_0$, the sum of the measures of $B_j$, $j_0 \le j \le N$, grows at least as a positive constant times $\log N$. By (6.8), the integral of the square of the sum of their indicators is $O((\log N)^2)$. Cauchy–Schwarz consequently gives a positive lower bound, independent of $j_0$, for

$$\left| \bigcup_{j \ge j_0} B_j \right|.$$

Continuity of measure along these decreasing tail unions shows that

$$|\limsup_{j\to\infty} B_j| > 0. \tag{6.9}$$

Together with (6.6), this proves the proposition. □

*Proof of Theorem 1(i).* Fix a nondegenerate compact interval $K_0 \subset (-1, 1)$, and suppose that it contains no point satisfying the assertion. Then

$$\lambda_n(x) - c_0 \log n \longrightarrow -\infty \qquad \text{for every } x \in K_0. \tag{6.10}$$

The closed sets

$$F_M = \{x \in K_0 : \lambda_n(x) \le c_0 \log n + M \text{ for every } n \ge 2\}, \qquad M = 1, 2, \dots,$$

cover $K_0$, by (6.10). Baire's theorem supplies a nondegenerate closed subinterval $I \subset K_0$ on which (4.6) holds with a fixed $C$. Proposition 9 contradicts (6.10). Thus every such $K_0$ contains a good point, proving density. □

# 7 Consequences

**Corollary 10** (Positive-measure sets)**.** *Let $E \subset (-1, 1)$ be measurable with $|E| > 0$, and let $0 < c < 2/\pi$. For all sufficiently large $n$, there is $x \in E$ such that $\lambda_n(x) > c \log n$.*

*Proof.* Otherwise $\lambda_n(y) \le c \log n$ for every $y \in E$ and every $n$ in an infinite set $S$. Repeating the proof of Theorem 1(ii) along $S$ gives the same contradiction $|E|/A \ge |E|$ with $A > 1$. □

# Code availability

The Lean 4 formalization is available at `https://github.com/FireflySentinel/erdos-1132`. Theorem 1 is `Erdos1132.theorem1` in `Erdos1132/Theorems.lean`. The two estimates quoted from Tao in Proposition 5 are also proved directly in Lean, with no additional axioms.

# Declaration of generative AI and AI-assisted technologies in the writing process

GPT-6 Astra proposed the derivative-jump and positive Cauchy-transform arguments, drafted the manuscript, and generated the Lean formalization. GPT-5.6 Sol and Claude Opus 5 were used to organize earlier research material and to review the Lean code. The author checked the arguments, completed the final manuscript, and takes full responsibility for the content.

## References


[1] S. Bernstein, Sur la limitation des valeurs d'un polynôme $P_n(x)$ de degré $n$ sur tout un segment par ses valeurs en $(n+1)$ points du segment, *Izv. Akad. Nauk SSSR*, Ser. VII (1931), no. 8, 1025–1050.

[2] T. F. Bloom, *Erdős Problem #1132*, `https://www.erdosproblems.com/1132`. Accessed 6 September 2026.

[3] L. Brutman, Lebesgue functions for polynomial interpolation—a survey, *Ann. Numer. Math.* **4** (1997), 111–127.

[4] P. Erdős, Problems and results on the theory of interpolation. II, *Acta Math. Acad. Sci. Hungar.* **12** (1961), 235–244.

[5] P. Erdős, Problems and results on the convergence and divergence properties of the Lagrange interpolation polynomials and some extremal problems, *Mathematica (Cluj)* **10** (1968), 65–73. `https://combinatorica.hu/~p_erdos/1967-20.pdf`.

[6] P. Erdős and P. Turán, An extremal problem in the theory of interpolation, *Acta Math. Acad. Sci. Hungar.* **12** (1961), 221–234.

[7] P. Erdős and P. Vértesi, On the almost everywhere divergence of Lagrange interpolatory polynomials for arbitrary system of nodes, *Acta Math. Acad. Sci. Hungar.* **36** (1980), 71–89.

[8] P. Erdős and P. Vértesi, On the Lebesgue function of interpolation, in *Functional Analysis and Approximation* (Oberwolfach, 1980), P. L. Butzer, B. Sz.-Nagy and E. Görlich (eds.), International Series of Numerical Mathematics **60**, Birkhäuser, Basel, 1981, 299–309. doi:10.1007/978-3-0348-9369-5_27.

[9] G. Faber, Über die interpolatorische Darstellung stetiger Funktionen, *Jahresber. Deutsch. Math.-Verein.* **23** (1914), 192–210.

[10] Q. Gu, *Nonuniform additive constants for Lebesgue functions*, preprint, 2026. `https://github.com/FireflySentinel/erdos-1132/blob/main/paper/NONUNIFORM.pdf`.

[11] T. Tao, *Local Bernstein theory, and lower bounds for Lebesgue constants*, arXiv:2603.21453v3 (2026).

[12] P. Vértesi, Optimal Lebesgue constant for Lagrange interpolation, *SIAM J. Numer. Anal.* **27** (1990), 1322–1331. doi:10.1137/0727075.